Swarnabha Ghosh

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Swarnabha Ghosh | swarnabhaghosh

Howrah, West Bengal - 711302, India

Objective

To leverage my knowledge in Artificial Intelligence and Machine Learning to develop scalable, data-driven solutions that solve real-world problems, while continuously learning and contributing to innovative projects in a dynamic tech environment.

Education

- **MCKV INSTITUTE OF ENGINEERING** — *September 2023 - Present*
  Bachelor of Technology in Computer Science — Howrah, West Bengal, India
  - Grade: 9.74/10.0 (Upto 5th Semester)
- **NEW ANDUL HIGHER CLASS SCHOOL** — *January 2015 - April 2023*
  Higher Secondary Education — Howrah, West Bengal, India
  - An aggregate of 89% in standard 12 and 95% in standard 10
  - Ardent participant of extra-curricular activities and competitions

Experience

- **Summer Intern – IEEE CS SBC, IIT Kharagpur** — *July 2025 - November 2025*
  Indian Institute of Technology (IIT), Kharagpur — Kharagpur, India
  - Formulated an Integer Linear Programming (ILP) model to solve the dynamic aircraft parking allocation problem, incorporating real-time revenue potential and operational constraints.
  - Developed an automated scheduling system that leverages the ILP model to optimize parking bay assignments, to generate maximum revenue.
  - Increased airport profitability and enhanced ground control efficiency by mitigating assignment conflicts and maximizing the utilization of high-value assets through a data-driven scheduling approach.

Projects

- **Movie Recommender System:** — *September 2025*
  Tools: Python, Pandas, NumPy, Scikit-learn, NLTK, ast, Streamlit, Pickle []
  - Developed a hybrid movie recommendation system using content-based filtering (genres, keywords) and cosine similarity with Scikit-learn's CountVectorizer on a 5,000+ movie dataset, integrated into a full-stack Streamlit web app that provides personalized recommendations with dynamically fetched posters via the TMDB API.
- **PlantIQ:** — *October 2025*
  Tools: Python, Pandas, Scikit-learn, TensorFlow, Keras, Numpy, PIL, Streamlit []
  - Developed a system that uses Random Forest classifier to recommend crops based on soil (pH, nutrients) and weather data (temperature, rainfall) and a MobileNetV2-based Convolutional Neural Network (CNN) for plant disease classification from leaf images, enabling fast and accurate detection of multiple disease types, enhancing agricultural yield and operational efficiency.
- **CivicLens – Smart Civic Issue Management System (Group Project):** — *October 2025*
  Tools: Next.js, Tailwind CSS, Express.js, FastAPI, MongoDB, Python, TensorFlow, Scikit-learn []
  - CivicLens is a full-stack web application with Citizen, Admin, and Worker dashboards, enabling users to report civic issues via text and images. The web development was done using Next.js, Tailwind CSS, Express, and MongoDB. I developed an AI module and FastAPI backend, integrating a MultiOutputClassifier for text-based issue classification and a CNN–RNN hybrid deep learning model (Conv2D + LSTM) for image classification to automate issue categorization and resolution management.

Memberships

- **Institution of Electronics and Telecommunication Engineers (IETE)** — *November 2025 - Present*
  Student Member

Skills

- **Programming Languages:** Python, Java, C, C++
- **Backend & Database:** Flask, FastAPI, SQL
- **Data Science & Machine Learning:** Pandas, NumPy, Scikit-Learn, TensorFlow, Matplotlib, Seaborn
- **Computer Vision & Image Processing:** OpenCV, Pillow, YOLO (Object Detection)
- **Version Control:** Git, GitHub
- **Mathematical & Statistical Tools:** NumPy, SciPy, PuLP
- **LLM Tools:** OpenAI, Gemini, Groq, Ollama, Hugging Face (transformers)